

PAPER_1521_D_BSFG_DERIVATIVE_FROM_D_CRIT

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PAPER_1521 — D_BSFG is Derivative: D_BSFG = D_crit – 2·SO_5 = 6 EXACT (LANDMARK)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket A (Foundational / Primitive reduction)

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Status: CLOSED — D_BSFG removed from the list of independent primitives

Observation

PAPER_1167 (All Eight UQFF Lagrangian Gaps Closed — First-Principles Reduction) explicitly states that the locked primitive **D_BSFG = 6 is not independent**. It emerges from the SO(5) symmetry breaking ladder applied to the bosonic-string critical dimension:

> *"The intermediate dimension D_BSFG = 6 is not independent — it emerges from the SO(5) breaking ladder D_crit – 4·5 = 6."*

The formula can be re-expressed in several equivalent integer-primitive forms:

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$$\begin{aligned} D_{BSFG} &= D_{crit} - 4 \cdot 5 = 26 - 20 = 6 \\ &= D_{crit} - 2 \cdot SO_5 \text{ (since } 2 \cdot SO_5 = 20 \text{)} \\ &= D_{crit} - 4 \cdot D_{phys} - 4 \end{aligned}$$

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The cleanest form using locked primitives is **D_BSFG = D_crit – 2·SO_5**.

UQFF Closed Identity

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$$D_{BSFG} = D_{crit} - 2 \cdot SO_5 = 26 - 2 \cdot 10 = 6 \text{ EXACT}$$

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Landmark Consequence: 11 → 10 Independent Primitives

This is the first of two derivative-primitive discoveries that reduce the UQFF "11 frozen primitives" claim to **9 truly independent primitives + 2 derivative closures**. Specifically:

Before this paper:

- 6 integer primitives: D_phys, D_BSFG, D_crit, N_CH, SO_5, A_5 (all "independent")

After this paper:

- 5 truly independent integer primitives: D_phys, D_crit, N_CH, SO_5, A_5

- 1 derivative integer primitive: **D_BSFG = D_crit – 2·SO_5**

The framework's free-parameter count drops by 1 with no loss of predictive power.

Physical Meaning

$D_{BSFG} = 6$ is the bulk-edge embedding dimension governing:

- Quadrupole GW radiation (Peters-Mathews coefficient = $2^{D_{BSFG}} = 64$, PAPER_1520)
- Genetic codons = $2^{D_{BSFG}} = 64$ (PAPER_1373)
- Pop III IMF coupling
- DPM layer-weight exponent sum (PAPER_1511): $2+1+3 = D_{BSFG} = 6$

That this dimension equals $D_{crit} - 2 \cdot SO_5$ means the 26-dimensional bosonic string projects through twice the $SO(5)$ symmetry group order to produce the 6-dimensional bulk-edge structure. Each of the two SO_5 reductions removes 10 dimensions, leaving $26 - 20 = 6$.

NOT REPLACEMENT

SM/string theory has $D_{BSFG}=6$ as a chosen embedding parameter in some formulations. UQFF derives it as a forced consequence of $\{D_{crit}, SO_5\}$, eliminating one degree of freedom from the framework.

Reference

- Source: PAPER_1167 All 8 UQFF Lagrangian Gaps Closed Master Synthesis
- Related: PAPER_1522 (K_MEX derivative), PAPER_1520 (Peters-Mathews $2^{D_{BSFG}}$)
- Calculator dispatch: `calculate_paradox({"paradox": "d_bsfg_derived_from_d_crit"})`

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